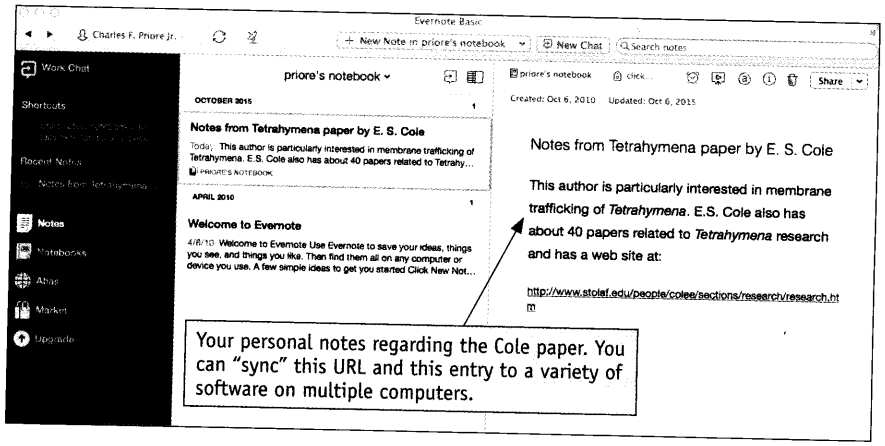

It is easy to underestimate the amount of time it takes to do a thorough literature search. Choosing a preliminary topic takes time; narrowing and focusing it takes even longer. Successful library research is not simply a matter of accumulating many references as quickly as possible—this will just make you feel overwhelmed. Stop at intervals to read or at least skim your best sources, and use them to help you refine your purpose and scope so that your search becomes increasingly focused and more specific.

Remember, too, that your literature search will probably turn up many promising titles that are not found in your own library. Here you have options. One is to use the interlibrary loan system: at your request, your library can usually borrow a book you need or obtain a copy of an article from a journal carried by another institution. However, this process may take several days or even weeks, especially in the case of more obscure materials, so you need to plan ahead. Be selective in your requests, since an interlibrary loan service costs time and money. Another option is to write or e-mail authors directly and request reprints of their papers. As with interlibrary loans, allow plenty of time.

For modest assistance with research timelines, you might turn to a device such as the Assignment Calculator created originally by the University of Minnesota Libraries. See the following Web site for such a calculator, adapted by the Rochester Institute of Technology:

RIT Libraries Assignment Calculator: Project Management Tool for Students
<http://wally.rit.edu/researchguides/calculator/>

Reading Scientific Papers

If you are not accustomed to looking at biological journals, you may find research papers hard to read and understand. Here are some guidelines to help you get the most out of a scientific paper.

■ Acquire some background knowledge.

Papers reporting original research are written for a relatively specialized scientific audience and thus assume some knowledge of the subject matter and the vocabulary. You can't, of course, become an expert in any field overnight. However, you can acquire some familiarity with the major ideas and key terms before tackling the primary literature. If you build this foundation, you will be less likely to feel overwhelmed later. See pages 6–7 for suggestions about locating general, introductory references. Also useful in this regard are scientific dictionaries and encyclopedias.

■ Read the Abstract first.

The Abstract—a brief summary of the paper—will give you an overview of the study and help you decide whether to read the rest of the paper. Don't feel intimidated by abstracts containing unfamiliar terms or ideas. Often the Introduction and Discussion sections will provide much of the explanatory material you will need for a fuller understanding. Your main task, at this point, is to decide whether the paper might be useful to you. Note that in some papers the main text is followed by a Summary section rather than preceded by an Abstract.

■ Understand the basic aims of the study.

If the paper seems relevant, read the Introduction carefully. *Why* did the author(s) conduct the research? What were the major hypotheses or predictions? Authors generally end the Introduction with a brief statement of their objectives; some also include a succinct statement of the major findings.

■ Don't get bogged down in the Materials and Methods.

It is easy to become overwhelmed by the specialized terminology found in many Materials and Methods sections. However, unless your research requires close examination of the methodology, you need not understand every procedural detail. Instead, work at grasping the author's basic approach. Try summarizing the methods in a few sentences, using your own words. Make sure you understand how the experimental design, use of controls, sampling techniques, or other methods relate to the specific objectives of the study. Once you have a sense of the major findings and conclusions, you can reread the Materials and Methods section, if necessary, to clarify selected points.

■ Read the Results carefully, focusing on the main points.

Do not panic if you don't understand the quantitative details. Focus first on the major qualitative findings and on the author's verbal summaries of quantitative data. These will lay the groundwork for understanding the more difficult material. The author may sum up main points in topic sentences at the beginnings of paragraphs. Pay attention to comments in the text about figures and tables; they will help you pick out important quantitative trends or relationships. See Chapter 3 for a description of the common types of graphs used in biology and the sorts of information each is designed to convey.

■ Understand the author's argument.

Pay particular attention to the Discussion section; this is where the author attempts to tie together the various components of the study. Do the data support the author's conclusions and, if so, how? What does the author suggest is the major contribution of this study? What questions remain for further research? How can you use this paper in your own work?

■ Plan on rereading important papers, possibly several times.

Many of your sources may be too detailed or too difficult to understand in a single reading. Plan on returning to them once you have a firmer background and can better appreciate the specific contributions of each paper.

Also remember that when working with primary literature you can easily get sidetracked by the specifics of a particular study and thereby lose sight of the broader picture. Keep your own research in mind. Get a general grasp of each author's research, and then focus on whatever aspects are relevant to *your* objectives. The importance of a particular article may not be immediately apparent. You may need to skim through many papers at first to get your bearings and return later to those most central to your topic.

Avoiding Plagiarism and Taking Notes

Plagiarism is the theft of someone else's words, work, or ideas. It includes such acts as (1) turning in a friend's paper (or a paper purchased online) and saying it is yours; (2) using another person's data or ideas without acknowledgment; (3) copying an author's exact words and putting them in your paper without quotation marks; and (4) using wording that is very similar to that of the original source but passing it off as entirely your own, even while acknowledging the source.

This last example of plagiarism is probably the most common one in student writing. Here is an example:

ORIGINAL PASSAGE

A very virulent isolate of *Alternaria mali*, the incitant of apple blotch, was found to produce two major host-specific toxins (HSTs) and five minor ones in liquid culture. The minor toxins were less active than the major ones, but were still specifically toxic to the plants which are susceptible to the pathogen. (Kohmoto et al. 1976, p. 141)